

These lecture notes are prepared by **Salman Ahmad**, who is both a **teacher** and a **web developer**. The purpose of this material is to make the topic clear and useful for students of Class 9 Computer Science.

Definition

Emerging technology refers to new, innovative, and rapidly developing technologies that have the potential to change industries, societies, and human life.

Examples: Artificial Intelligence (AI), 5G Networks, Robotics.

Artificial Intelligence (AI)

Introduction

Artificial Intelligence (AI) refers to the ability of machines or computer systems to think, learn, and make decisions like humans. AI is widely used today. From self-driving cars, chatbots, and medical diagnosis to gaming and e-commerce recommendations.

Historical Context of AI

- 1955: First AI program Logic Theorist (by Allen Newell & Herbert A. Simon).
- 1956: Term “Artificial Intelligence” introduced by John McCarthy at the Dartmouth Conference.
- 1950s–1960s: Early AI research → problem-solving and symbolic logic.
- 1970s–1980s: Expert systems (computers making decisions like humans).
- 1990s: Machine learning → computers started learning from data.
- 2000s: Growth of deep learning, robotics, and natural language processing (NLP).
- 2011s: Voice assistants like Siri, Alexa.
- 2023–Present: AI models like ChatGPT that understand and generate human-like text.

Applications of AI Healthcare

Definition: Use of AI in medical field for diagnosis, treatment, and patient care.

- AI detects cancer in X-rays faster than human doctors.
- AI-powered robots assist in surgery.
- Wearable devices use AI to monitor heart rate and alert patients.

Education

Definition: AI in education helps improve learning, teaching, and management.

- AI chatbots answer student questions online.

Transportation

Definition: AI is used in self-driving cars and traffic systems to make roads safer and reduce traffic. It helps control traffic lights and avoid accidents.

- Tesla's self-driving cars.
- GPS apps using AI to suggest fastest routes.

Automobile

Definition: AI is improving cars by adding smart features like automatic braking, lane assist, and better performance. It also helps in vehicle maintenance and makes driving safer and more comfortable.

Example:

Many modern cars from companies like BMW use AI for parking assistance and warning systems to avoid accidents.

Finance

Definition: AI in finance automates transactions, fraud detection, and customer service.

- Banks use AI chatbots for customer queries.
- AI detects credit card fraud.
- AI advisors suggest investment plans.

Gaming

Definition: AI controls characters, adjusts game difficulty, and makes games more interactive.

- AI adapts difficulty based on player's skill.

Social Media

Definition: AI makes social platforms smarter and personalized.

- YouTube/TikTok recommending videos.
- Facebook detecting fake news.
- Netflix suggesting shows based on watching history.

Agriculture

Definition: AI in farming helps improve crop quality and productivity.

- AI drones check crop health.
- AI predicts best planting/harvesting times.
- Automated irrigation saves water.

E-commerce

Definition: AI improves shopping and business operations.

Amazon product recommendations. Chatbots helping customers.
AI fraud detection in online payments.